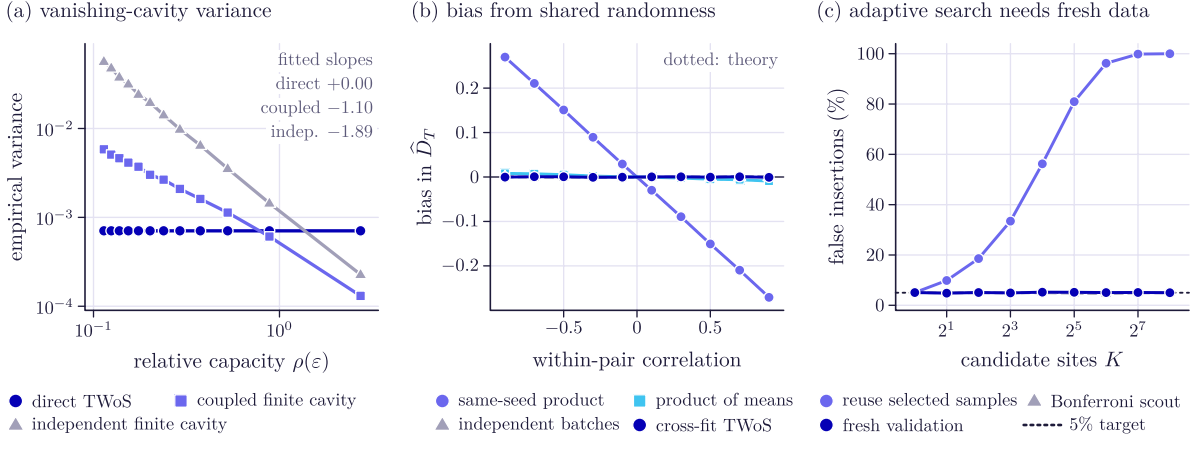

(a) vanishing-cavity variance
fitted slopes
direct +0.00
coupled −1.10
indep. −1.89
empirical variance
10^{-2}
10^{-3}
10^{-4}
10^{-1}
10^{0}
relative capacity $\rho(\varepsilon)$
direct TWoS
coupled finite cavity
independent finite cavity
(b) bias from shared randomness
dotted: theory
bias in $\widehat{D}_T$
0.2
0.1
0
−0.1
−0.2
−0.5
0
0.5
within-pair correlation
same-seed product
product of means
independent batches
cross-fit TWoS
(c) adaptive search needs fresh data
false insertions (%)
100
80
60
40
20
0
2^1
2^3
2^5
2^7
candidate sites K
reuse selected samples
Bonferroni scout
fresh validation
5% target